

Kingdom of Saudi Arabia
Ministry of Education
Shaqra University
College of Computing and Information Technology
Computer Science Department



MANSAK AMIN

منسك آمن

ML-DL project

2024-2025

Supervisor:

Dr. Samia Dardouri

Students:

Meteabh Aldawsri
Hanan Othman
Shoroq ALotibi
Nouf ALqasim
Deema Alsaad

ID:

443500038
443500118
443500110
443500113
444500629

Abstract

This project introduces a smart AI-based framework for real-time crowd monitoring and management during Hajj and Umrah. Using deep learning object detection (YOLOv8) and live video analytics via computer vision, the system counts people and triggers alerts upon overcrowding. It provides visual dashboards and detection accuracy metrics, enabling better decision-making and fast intervention.

موجز المشروع باللغة العربية

مع تزايد أعداد الحجاج والمعتمرين سنوياً، تظهر الحاجة الملحة إلى حلول ذكية تساعد في تنظيم وإدارة الحشود بشكل يضمن سلامة ضيوف الرحمن. يهدف مشروع "منسك آمن" إلى استخدام تقنيات الذكاء الاصطناعي لرصد الكثافات البشرية والتنبيه عند حدوث ازدحام مفرط في المواقع الحساسة مثل صحن الطواف والمسعى.

Table of Contents

Topic	page
1. Introduction	4
1.2 Problem description	4
1.3 Objective	4
1.4 Digitization and vision 2030	4
2. Literature Review	5
3. Requirements	6
3.1 Hardware	6
3.2 System& Software	6
3.3 Functional Requirements	6
3.4 Non-Functional Requirements	7
3.5 Admin Level	8
3.6 Data and contents	9
4. Project Management Strategies Involved in the Project	10
5. Materials and Methods	11
5.10 System Integration and Interface Design	12
6. Results and Discussion	18
7. Conclusions and Future Work	20
References	21

1. Introduction

Hajj is one of the largest annual human gatherings in the world, involving millions of pilgrims converging in a limited geographical area. Managing the movement of such massive crowds poses significant challenges for ensuring safety and efficiency. This project aims to leverage artificial intelligence technologies, particularly computer vision and deep learning, to develop a smart system capable of analyzing crowd density, movement patterns, and predicting potential congestion or dangerous crowd behaviors. Our proposed system, "Mansak Amin," serves as a preventive safety tool during Hajj by providing real-time monitoring, early warnings, and decision support to authorities. It will not function as a standalone tool, but will be integrated with the entire network of surveillance cameras across all areas of the Grand Mosque and surrounding zones, enabling comprehensive oversight and coordinated response mechanisms. Furthermore, future extensions of the system will include integration with drone technologies to allow faster aerial monitoring and assistance. Additional features will enable rapid identification of missing individuals through image input; by uploading a photo, the system will analyze camera feeds across the Haram to locate and report the individual's last known position in real-time.

1.2 Problem Description

Managing large-scale pilgrim crowds during Hajj presents a complex and multifaceted problem. With millions of individuals moving simultaneously through sacred sites such as the Grand Mosque, Mina, Arafat, and Muzdalifah, the potential for overcrowding, stampedes, heat-related illnesses, and separation from groups is high. Traditional crowd management relies heavily on human surveillance, physical barriers, and static monitoring systems, which are often reactive rather than preventive. These methods face critical limitations in detecting subtle changes in crowd dynamics, predicting dangerous conditions, and responding swiftly across vast areas.

Moreover, the inability to quickly locate lost individuals in such densely populated areas increases the emotional distress for pilgrims and the operational burden on authorities. The need for real-time, intelligent, and scalable solutions has never been more urgent. While CCTV infrastructure exists across the Haram, there is a lack of an integrated AI-powered platform that can proactively interpret video feeds, detect anomalies, and assist in decision-making.

The proposed "Mansak Amin" system addresses these challenges by offering a hybrid AI framework capable of real-time analysis, predictive modeling, and facial recognition. It will not only monitor crowd density and movement, but also, in future phases, support missing person detection via photo input and integrate with drones for enhanced coverage and responsiveness.

1.3 Objective: Develop an AI-powered system for real-time crowd monitoring during Hajj, integrated with mosque cameras, with future support for drones and missing person detection.

1.4 Digitization and Vision 2030

Saudi Arabia's Vision 2030 emphasizes digital transformation across all sectors, especially in services provided to pilgrims during Hajj and Umrah. One of the core goals is to enhance the safety, efficiency, and satisfaction of pilgrims through smart technologies and AI-driven solutions. The "Mansak Amin" project aligns directly with this vision by offering a scalable, intelligent surveillance and crowd management system powered by artificial intelligence. Through integration with the Grand Mosque's camera infrastructure and future drone capabilities, the system supports the Kingdom's ambition to become a global leader in smart city and crowd management technologies, while ensuring a safer and more seamless pilgrimage experience.

2. Literature Review

The rapid growth in the number of pilgrims during Hajj has underscored the urgent need for intelligent crowd management and smart guidance systems to ensure safety, efficiency, and comfort. Early solutions, which relied primarily on basic surveillance and manual control, proved inadequate in addressing the scale and dynamic nature of pilgrim movement. To overcome these challenges, recent research has turned to advanced technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), and computer vision to enhance crowd analysis, navigation, and risk mitigation [1], [2].

AI-powered systems now analyze real-time data from surveillance cameras, thermal sensors, and mobile devices to predict congestion hotspots and optimize crowd flow. For example, Alghamdi and Mehmood (2023) utilized edge AI and mobility data to monitor and manage pilgrim movement with minimal latency [1]. Similarly, Alharthi and Alfarraj (2024) introduced an IoT-enabled system that integrates sensor data with predictive analytics for real-time crowd control, enabling early anomaly detection and prevention of stampedes [2].

In addition to safety, smart guidance systems have been implemented to support navigation and enhance the pilgrim experience. Technologies such as BLE beacons, GPS, and QR-code-based mobile apps provide location-based services, offering real-time directions and personalized alerts. According to the Saudi Ministry of Media, virtual platforms and intelligent signage significantly improved navigation during Hajj 2024 [3]. ITAS further reported that AI-driven sensor networks and smart applications contributed to more effective crowd management and quicker incident response times [4].

Moreover, robotic assistants and facial recognition technologies have been deployed across holy sites to streamline access control, respond to pilgrim inquiries, and monitor health conditions. CNN Business Arabic highlighted the introduction of AI-enabled robots, intelligent surveillance, and multilingual support systems during Hajj 2024 as critical components in enhancing safety and coordination [5].

Despite these advancements, current implementations often operate in isolation—focusing on specific components such as detection or guidance—without offering an integrated, adaptive solution. Our proposed model addresses this gap by unifying real-time crowd sensing, dynamic route recommendation, and multilingual smart guidance into a single platform. Leveraging edge computing, AI-driven predictions, and contextual alerts, this system provides a comprehensive solution for managing the complexities of modern-day pilgrimage.

3. Requirements

3.1 Hardware

The implementation of the system relies on advanced hardware environments that support rapid and efficient data processing and analysis. High-performance devices have been selected to meet the complex technical requirements of the system. The key hardware components used in the system include:

1. **Surveillance Cameras:** A set of high-definition surveillance cameras has been installed in strategic locations within the Holy Mosque and its surrounding areas. These cameras are the primary source of visual data needed for the system to analyze crowd movement. Cameras equipped with night vision technology are selected to ensure clear images in all environmental conditions, including low-light scenarios.
2. **Servers:** The connected servers used for data processing are equipped with powerful processors capable of handling large amounts of data collected from surveillance cameras. Multi-core processors and large Random Access Memory (RAM) are used to ensure the system can process data in real time. Cloud servers are also utilized for scalability, handling the growing volume of data and expanding system operations in the future.
3. **Graphics Processing Units (GPU):** To accelerate image and video processing tasks, Graphics Processing Units (GPUs) are utilized, which support deep learning applications. These GPUs assist in speeding up the training of intelligent models, such as Deep Neural Networks (DNN) and LSTM-based models, improving the overall system performance.
4. **Unmanned Aerial Vehicles (Drones):** In the future, drones will be integrated into the system to provide aerial monitoring. Drones are equipped with high-definition cameras and advanced sensors to capture images and videos from different angles. Drones contribute additional data on crowd movement from an aerial perspective, enhancing overall analysis and predicting congestion areas more accurately.
5. **Workstations:** High-performance workstations are equipped to support the monitoring and analysis teams. These workstations are outfitted with multiple display screens that allow operators to monitor live data and analyses in real time. They are also equipped with specialized software for data analysis and video processing.

3.2 System & Software

The implementation of the intelligent system for crowd management during Hajj relies on a combination of software tools and systems that ensure high performance and fast processing and analysis capabilities. These software components use advanced technologies that support the intelligent analysis of visual data and crowd movement. The key software tools used in the system are:

1. **Programming Language Python:** Python is one of the core programming languages used in the system due to its high capability to handle large datasets and its flexibility in developing complex algorithms. It is used to develop software that processes and analyzes the data input from surveillance cameras. Python also supports specialized libraries for image and video processing, such as OpenCV and Pillow, and facilitates the development of intelligent models using deep learning libraries such as TensorFlow and Keras.
2. **OpenCV:** OpenCV is one of the essential tools for image and video processing. It is used to extract features from visual data, analyze motion, and predict congestion or hazardous behaviors in real-time. OpenCV also enables image enhancement and noise reduction, contributing to improved analysis accuracy.
3. **TensorFlow and Keras:** TensorFlow and Keras are used to build and train deep learning models that rely on artificial neural networks. These models process and analyze temporal data related to crowd movement using advanced techniques such as Long Short-Term Memory (LSTM), which helps predict congestion areas or potential accidents based on historical and real-time data.
4. **Apache Kafka:** Apache Kafka is used as a platform for managing real-time data streams. This platform collects and continuously analyzes data from the surveillance cameras in the Holy Mosque and surrounding areas, ensuring rapid and accurate data updates. Data is sent to the servers for real-time analysis.
5. **Database Management System (DBMS):** Database Management Systems like PostgreSQL are used to store and analyze the data generated by the surveillance cameras. The DBMS helps manage large volumes of data, ensuring easy access and efficient updates. Data is stored in an organized database to speed up retrieval and accurate analysis.
6. **Cloud Systems:** To enhance the scalability and capacity of the system, cloud platforms like Amazon Web Services (AWS) or Google Cloud Platform (GCP) are used. These platforms enable hosting servers and processing data on a large scale, improving performance and ensuring service continuity.
7. **Unmanned Aerial Vehicle (UAV) Systems:** In the future, specialized software for controlling drones (UAVs) will be used to enable the system to provide real-time aerial monitoring. Drones will be integrated with tracking and data analysis systems to provide additional oversight of the crowds.

3.3 Functional Requirements

1. Crowd Detection and Tracking

- **Real-Time Detection:** The system must continuously monitor the video feed and detect the presence of people in real-time using **YOLOv8** or other object detection models.
- **Individual Tracking:** The system must be able to track individual people across video frames, assigning unique identifiers to each person and maintaining their movement history.
- **Overcrowding Detection:** The system must identify when the crowd density exceeds a pre-set threshold (e.g., 60 people) and classify the area as "overcrowded" or "normal."

2. Person Identification (Missing Person Detection)

- **Image Upload for Missing Persons:** The system should allow users to upload images of missing persons through the user interface.
- **Person Search and Alert:** The system should be able to compare the uploaded image with the live video feed and alert operators if a match is found.
- **Notification System:** When a person is identified or located, the system should send notifications or alerts to relevant personnel.

3. Crowd Management Alerts and Notifications

- **Real-Time Alerts:** The system must generate alerts when overcrowding is detected. Alerts should be both visual (e.g., a flashing message on the display) and auditory (e.g., a sound alarm).
- **Audio Notifications:** An audio warning should be played when a certain crowd threshold is reached, helping notify nearby personnel or individuals of the situation.
- **Threshold Customization:** The threshold for overcrowding must be customizable by the system administrators based on varying crowd management requirements.

4. User Interface (UI) and Visualization

- **Dashboard Interface:** The system should provide an intuitive dashboard where operators can see the real-time number of people in a given area, the status of overcrowding, and detection accuracy.
- **Real-Time Data:** The dashboard must display the current count of people, time elapsed, and detection accuracy in real-time, along with a status indicator (e.g., "Overcrowded" or "Normal").
- **Graphical Data Representation:** The system should generate graphical representations of crowd trends over time (e.g., a line chart showing the number of people per minute).
- **Live Video Stream:** The system should allow operators to view the live video stream along with overlaid information such as the number of people, status, and tracking IDs.

5. Data Storage and Session Management

- **Session Data Logging:** The system must log all session data including timestamps, the number of people in view, and detection accuracy over time. This data can be used for analysis and reporting.
- **Historical Data Access:** Operators should be able to review historical data for trend analysis and incident investigation.
- **Temporary Data Storage:** For uploaded video files and images, the system must provide temporary storage that deletes data after processing to comply with privacy and security standards.

6. Real-Time Video Processing

- **Video Feed Integration:** The system must support the integration of various video sources, including uploaded video files or live streams from external cameras (webcams, drones, etc.).
- **Video Processing in Real-Time:** The system must be capable of processing video feeds and performing person detection, tracking, and status updates with minimal latency.

7. Performance and Scalability

- **Low Latency:** The system should minimize latency in detecting and tracking people, providing near-instantaneous feedback to operators.
- **Scalability:** The system must be scalable to handle multiple video feeds from various cameras across the mosque or other high-traffic areas, ensuring smooth operation as the scale of monitoring increases.

8. Security and Privacy

- **Data Encryption:** All sensitive data, such as personal images or video footage, should be encrypted to ensure privacy and security.
- **Access Control:** The system must have robust access controls to ensure that only authorized personnel can access the dashboard and sensitive information.
- **Compliance with Local Regulations:** The system should comply with relevant data protection regulations, especially concerning video surveillance and personal data usage during Hajj.

9. Sound and Visual Alerts

- **Alert Customization:** Alerts should be customizable, including options for different types of audio alerts or visual indicators based on the severity of the crowd situation.
- **Alert Time Control:** The system should ensure that alerts are timed correctly to avoid unnecessary repetition. For example, a sound alert should be triggered only once when overcrowding is detected.
- **Emergency Response Coordination:** In case of overcrowding, the system should allow operators to quickly coordinate with security or emergency response teams through alerts and integrated communication tools.

10. System Monitoring and Maintenance

- **Error Detection:** The system must be able to detect errors, such as camera disconnection, system overload, or detection failures, and notify operators immediately.
- **Real-Time Health Monitoring:** The system should include health checks that monitor the status of all components (e.g., cameras, video processing units) to ensure continuous operation.

3.4 Non-functional Requirements

1. Performance

- **Real-Time Response:** The system must be capable of processing video data in real-time with minimal delay. Person detection, tracking, and congestion updates should occur within seconds after receiving the video data.
- **Handling Large Data:** The system should be able to process and analyze large amounts of data transmitted from multiple cameras or live streaming sources without significantly impacting performance.
- **Scalability:** The system should be scalable to handle an increasing number of cameras or locations in high-traffic areas without degrading overall performance.

2. Availability

- **Continuous Operation:** The system should be available throughout the operational period (24 hours a day during the Hajj season) with minimal downtime.
- **Redundancy and Backup:** The system should include mechanisms to ensure redundancy (such as live data backups) to prevent data loss in case of a system failure.

3. Security

- **Data Protection:** Sensitive data (such as video footage and individual images) must be encrypted both in transit and at rest to ensure privacy and prevent unauthorized access.
- **Access Control:** The system should have a mechanism for defining different access levels for authorized personnel, allowing control over who can view or modify data.
- **Privacy Compliance:** The system must comply with local and international laws and regulations regarding privacy and data protection (such as personal data protection laws).

4. Reliability

- **Automatic Fault Detection:** The system should have the capability to automatically detect failures in components (e.g., camera malfunctions or video processing issues) and send instant alerts to operators.
- **System Recovery Procedures:** In the event of a failure, the system must be capable of quickly restoring functionality from backups or the last valid state to ensure service continuity.

- **Performance Monitoring:** The system should continuously monitor performance and provide early warnings if a problem arises that may affect service.

5. Usability

- **Ease of Interaction:** The user interface should be intuitive and easy to use, allowing operators to interact with the system efficiently even under high-pressure situations.
- **Multi-Device Accessibility:** The system must be accessible through multiple devices, such as desktops, tablets, and smartphones, with user interfaces optimized for each device type.
- **Training and Support:** The system should include training programs for operators to ensure effective use of the system, and technical support should be available in case issues arise.

6. Maintainability

- **Regular Updates:** The system should allow for regular updates to ensure compatibility with the latest technologies and best practices in AI, as well as address any security vulnerabilities.
- **Comprehensive Documentation:** The system should have comprehensive documentation, including user manuals, troubleshooting guides, and standard operating procedures, to facilitate maintenance by technical teams.
- **Self-Diagnostics:** The system should have built-in self-diagnostic tools to help quickly identify technical issues, reducing the time required for repairs.

7. Flexibility

- **Horizontal Scalability:** The system should support easy horizontal scaling to handle increased numbers of cameras or locations without negatively affecting performance.
- **Integration with Other Systems:** The system should be flexible enough to integrate with other crowd management or security applications, such as alarm systems, mobile applications, or data analytics platforms.

8. Compatibility

- **Support for Various Devices:** The system should support a wide range of devices, such as different surveillance cameras, computers, and mobile phones, ensuring compatibility with any equipment already in use at the site.
- **Software Integration:** The system should be capable of interacting and integrating with other software tools, such as data analytics platforms, alert systems, or security management systems.

9. Environmental Sustainability

- **Low Energy Consumption:** The system should be designed to consume minimal energy, especially when deployed in large environments with numerous devices.
- **Electronic Waste Minimization:** The system should include policies for recycling equipment or minimizing electronic waste, if applicable.

3.5 Admin Level

- **Administrator:**

- **Responsibilities:** Manage user accounts, system settings, updates, and reports.
- **Permissions:** Full system access, including settings, user data, and advanced diagnostics.

- **Operator:**

- **Responsibilities:** Monitor real-time crowd data, respond to alerts, and manage crowd-related issues.
- **Permissions:** Access to live video, alerts, and limited system settings for crowd management.

- **Security Personnel:**

- **Responsibilities:** Use system data for crowd control and emergency responses.
- **Permissions:** View live data, respond to alerts, and coordinate with other teams.

- **Maintenance & Support Staff:**

- **Responsibilities:** Maintain system infrastructure, diagnose technical issues, and ensure system uptime.
- **Permissions:** Access to diagnostic tools, limited software settings.

- **Guest User:**

- **Responsibilities:** View crowd data and alerts.
- **Permissions:** Read-only access to live statistics and alerts.

- **External Integrators (API Users):**

- **Responsibilities:** Integrate external systems with **Manasik Safe** via API for data sharing.
- **Permissions:** Limited API access to crowd data and alerts.

3.6 Data and Contents

3.4.1 Data:

The data required for the system's operation is multifaceted and includes real-time, historical, and external data sources. These data types are essential to ensure accurate monitoring, analysis, and prediction of crowd behavior, as well as the identification of potential risks during the Hajj season.

The types of data that will be collected and processed by the system include:

- **Real-time Video Data:**
 - **Source:** Surveillance cameras across the Grand Mosque and surrounding areas.
 - **Purpose:** To monitor crowd density, movement patterns, and detect risky behaviors such as overcrowding or stampedes.
 - **Format:** Continuous video streams in high-definition (HD) format, with the ability to process footage at different time intervals.
- **Historical Crowd Movement Data:**
 - **Source:** Previous Hajj seasons and data collected from sensors or other crowd management systems.
 - **Purpose:** To analyze crowd behavior patterns and trends, identify high-risk areas, and predict potential congestion points.
 - **Format:** Structured datasets of crowd density, movement patterns, and incident reports.
- **Weather Data:**
 - **Source:** Real-time weather services, including temperature, humidity, wind speed, and precipitation data.
 - **Purpose:** To understand the impact of weather on crowd behavior (e.g., increased heat or rain leading to changes in crowd movement).
 - **Format:** Time-stamped weather data, including forecasts and real-time updates.
- **Prayer Times Data:**
 - **Source:** Islamic prayer time schedules.
 - **Purpose:** To account for prayer schedules which could influence crowd movements, especially at prayer times when crowds may congregate at specific locations.
 - **Format:** Time-based data aligned with daily prayer times.
- **Missing Person Data:**
 - **Source:** Photos of missing individuals uploaded by authorities or pilgrims.
 - **Purpose:** To track missing persons based on facial recognition or image matching with surveillance camera feeds.
 - **Format:** Image files (JPEG, PNG, etc.) and metadata related to the missing person's description.

The data will be processed in real-time and archived for future analysis and training purposes. The combination of these diverse data sources will allow the system to provide accurate predictions, alerts, and insights to ensure the safety and efficiency of the Hajj pilgrimage.

3.4.2 Contents:

The contents of the system refer to the various elements, resources, and materials that will be used within the platform to support its functionality. These include multimedia files, documentation, algorithms, models, and other essential resources that are integral to the system's operation. The contents will be categorized as follows:

1. Surveillance Camera Feeds:

- **Description:** Live video streams from surveillance cameras located throughout the Grand Mosque and its surrounding areas. These feeds will provide real-time visual data for monitoring crowd movement, density, and behaviors.
- **Format:** HD video streams, potentially in formats like RTSP (Real-Time Streaming Protocol) or HTTP Live Streaming (HLS).
- **Purpose:** To provide the system with the raw video data needed for processing, analysis, and decision-making.

2. Crowd Movement and Density Data:

- **Description:** Data regarding the movement patterns and density of crowds, derived from analysis of the video feeds and sensor-based systems deployed in and around the Hajj areas.
- **Format:** Structured data, including time-stamped density levels, crowd distribution, and movement vectors.
- **Purpose:** To enable the system to predict congestion points, detect crowd bottlenecks, and analyze crowd dynamics over time.

3. Machine Learning Models and Algorithms:

- **Description:** Pre-trained models and custom algorithms developed for crowd analysis, anomaly detection, and predictive analytics. These models are based on deep learning techniques and computer vision.
- **Format:** Files in formats such as TensorFlow (.pb), PyTorch (.pth), or ONNX (.onnx).
- **Purpose:** To process the video feeds, identify patterns, predict risks, and generate actionable insights (e.g., early warnings of crowd dangers).

4. Weather Data and Forecasts:

- **Description:** Data related to weather conditions that could influence crowd behavior during Hajj, including real-time forecasts, temperature, humidity, and precipitation.
- **Format:** Time-stamped weather data, provided in API formats such as JSON or XML.
- **Purpose:** To help the system assess how environmental conditions could affect crowd movement and safety.

5. Prayer Times Data:

- **Description:** Data representing prayer times, which could influence crowd movements during specific times of the day.
- **Format:** Structured data with timestamps aligned with prayer times (e.g., Fajr, Dhuhr, Asr, Maghrib, Isha).
- **Purpose:** To allow the system to factor in prayer schedules, predicting crowd surges during prayer times and enabling better crowd control.

6. Missing Person Database:

- **Description:** A database that contains images and descriptions of missing individuals, allowing for facial recognition or image matching with surveillance footage.
 - **Format:** Image files (e.g., JPEG, PNG) and associated metadata (e.g., name, age, description).
 - **Purpose:** To help locate missing persons by cross-referencing uploaded photos with live camera feeds.
7. **Reports and Logs:**
- **Description:** Data logs, incident reports, and alerts generated by the system's analysis of crowd behaviors, events, and potential risks.
 - **Format:** Structured logs (e.g., JSON, CSV), alerts, and reports in PDF or text format.
 - **Purpose:** To provide authorities with actionable insights, help with incident management, and document system performance.
8. **User Interaction Data:**
- **Description:** Data generated from users interacting with the system, including inputs for crowd-related queries, missing person searches, and system feedback.
 - **Format:** Logs of interactions, input data (e.g., search queries, feedback), and results.
 - **Purpose:** To improve user experience, track system usage, and fine-tune system performance over time.

These contents are critical for ensuring the system's accuracy, responsiveness, and reliability in real-time crowd management, safety monitoring, and predictive analysis during the Hajj pilgrimage.

4. Project Management Strategies Involved in the Project

4.1 Project Scope

The project scope defines the boundaries and deliverables of the "Mansak Amin" system, specifying what is included and excluded from the project. It outlines the high-level objectives, components, and functionalities that the system will achieve and deliver. The scope of the project encompasses the following key elements:

1. System Development:

- Development of an AI-based crowd management system to monitor, analyze, and predict crowd behavior during Hajj.
- Integration of real-time video streams from surveillance cameras to gather and process data.
- Implementation of machine learning models for crowd density analysis, movement tracking, and anomaly detection.
- Development of features for identifying missing persons using facial recognition or image matching.

2. Technology Stack:

- Use of computer vision and deep learning technologies to process video feeds and identify critical events in real-time.
- Integration with existing surveillance camera networks across the Grand Mosque and surrounding areas.
- Development of user interfaces for authorities to view real-time data, get alerts, and manage crowd safety measures.

3. Data Integration:

- Integration of live data streams, including surveillance camera feeds, weather data, prayer time schedules, and crowd density data.
- Development of algorithms for crowd prediction, anomaly detection, and early warning systems.

4. System Features:

- Real-time monitoring and analysis of crowd behavior.
- Prediction of congestion points and hazardous crowd conditions.
- Early warning system to alert authorities about potential risks.
- Missing person identification system integrated with real-time camera feeds.
- User interface for managing the system, accessing reports, and monitoring data.

5. System Deployment:

- Installation and configuration of the system across the Grand Mosque and Hajj-related areas.
- Integration with drone technologies for aerial surveillance (future extension).
- Regular testing and optimization of the system to ensure real-time performance and reliability.

6. Exclusions:

- The project will not cover the development of new surveillance camera infrastructure.
- The system will not directly manage crowd control actions but will provide data and insights to authorities.

- Integration with non-Grand Mosque areas or other pilgrim locations outside the scope of the project.

By clearly defining the project scope, the "Mansak Amin" system ensures that all stakeholders understand the project's objectives, limits, and deliverables. The goal is to enhance crowd safety and management during Hajj through real-time analysis, prediction, and intervention.

4.2 Role of each project team member (if any):

- Hanan Othman: Literature Review/References - User Level - Functional Requirements - business modal - Activity diagram.
- Meteabh Aldawsri: project Scope- Project Management - Future work and conclusion - Sequence diagram.
- Shoroq Alotaibi: Abstract - Digitization and Vision 2030 - Non-Functional Requirements - Results and Discussion - ER diagram.
- Nouf Alqasim: Introduction - Data and Contents - Design Overview – DFD diagram.
- Deema Alsaad: Problem – Objectives - Materials and Methods - Use Case diagram and description.

4.3 Time Plan of the Project

The Gantt chart describes:

A horizontal bar chart used in project management to visually represent a project plan over time.

Tasks	Jan				Mar				Apr			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Planning Phase:												
Choose the topic of the project												
Search about the topic												
Chapter 1: Introduction												
Write the background												
Write the problem and motivation												
Write digitization and vision 2030												
Chapter 2: Literature review												
Write the Literature review												
Analysis phase:												
Chapter 3: Requirements												
Write the hardware and Software												
Write the user level and data												
Chapter 4: Project Management Strategies												
Design phase:												
Chapter 5: Materials and Methods												
System Integration and Interface Design												
Results and Discussion												
Conclusion and Future work												

Table 1, Timeline of the project (term1)

4.4 Suggested Business Modal

The proposed business model of the "Mansak Amin" system is designed with a strong focus on national benefit, long-term sustainability, and applicability. It serves as a strategic solution aligned with Saudi Arabia's Vision 2030, particularly the goal of enhancing the safety and experience of pilgrims through smart technologies.

1. Government Partnership Model (Primary Focus):

- **National Implementation:** The system is intended to be adopted and maintained by Saudi government agencies (Ministry of Hajj, Public Security, Presidency of the Two Holy Mosques).
- **Strategic Investment:** Rather than a commercial product, it is positioned as a *national infrastructure* investment that directly supports crowd safety, emergency response, and digital transformation.
- **Budget Allocation:** Funding can come from the national budget allocated for Hajj operations and smart city initiatives.

2. Scalability & National Service Impact:

- **Phased Deployment:** The system can be deployed in stages (e.g., Grand Mosque, Mina, Arafat), ensuring gradual expansion and continuous evaluation.
- **Extensibility:** Future extensions could support Umrah seasons and other high-density areas (e.g., Masjid an-Nabawi).
- **Public Benefit:** Helps reduce casualties, enhance operational decision-making, and provide a proactive safety net during critical religious events.

3. Technology Transfer & Local Talent Involvement:

- Development and maintenance involve local tech companies and universities, promoting national innovation and creating job opportunities for Saudi engineers, data scientists, and AI specialists.

4. Supporting National Security and Planning:

- By integrating real-time analytics with emergency services, the system enhances national preparedness and response strategies.
- Data collected can inform long-term infrastructure planning and pilgrim flow optimization.

5. Optional Commercial Use (Future Opportunity):

- After successful national deployment, the system can be repackaged for international use during mass gatherings (e.g., Olympics, global religious events), generating revenue that supports its local operation.

5. Materials and Methods

5.1 Operating System

The "Mansak Amin" system is designed to be platform-independent, leveraging cloud-based infrastructure while supporting local deployments on Windows-based servers for real-time video processing and AI inference.

5.2 Documentation

Comprehensive documentation is a critical part of the "Mansak Amin" project to ensure clarity, maintainability, and ease of collaboration among team members and stakeholders. The documentation includes the following components:

- **System Architecture Documentation:** Details the design of system components, data flow, and interactions between modules.
- **Installation & Deployment Guides:** Step-by-step instructions for setting up the environment, installing dependencies, and deploying the system locally or on the cloud.
- **User Manuals:** Provides guidance for authorities and operators on how to use the system interface, access reports, and respond to alerts.
- **API Documentation:** Explains how to interact with system components programmatically through APIs, useful for integration with other systems.
- **Developer Documentation:** Includes code structure, standards, and guidelines for contributors and future development teams.
- **Security Documentation:** Outlines protocols for data protection, user access control, and system authentication procedures.

All documentation will be updated regularly and made accessible through a centralized, version-controlled repository (e.g., GitHub or GitLab).

5.3 UML Designing

Unified Modeling Language (UML) is a graphical language for visualizing, specifying, constructing, and documenting the artifacts of software systems. UML diagrams can be used to model the different aspects of a Mansak Amin Ai, such as the class diagram, use case diagram, sequence diagram, and activity diagram.

5.4 Graphics and Designs

This section highlights the visual and user interface design elements of the "Mansak Amin" system. Clear, intuitive, and responsive design is crucial for ensuring that authorities can interact with the system efficiently, especially in high-pressure scenarios during Hajj.

Key design components include:

1. User Interface (UI) Design:

- Dashboard for real-time camera feed visualization.
- Heatmaps for crowd density representation.
- Notification panel for anomaly and congestion alerts.
- Search interface for missing person identification via image input.
- Admin panel for system controls and data management.

2. Color Scheme and Theme:

- Professional and minimal color palette (e.g., blue, gray, white) for clarity and focus.
- High contrast UI elements to support visibility in all lighting conditions.

3. Icons and Visual Elements:

- Standardized icons for navigation, alerts, and camera status.
- Visual representation of crowd movement and risk zones using gradients.

4. Responsive Design:

- Optimized layouts for large control screens and tablets used by authorities.
- Scalable design to support future integration with mobile apps or drones.

5.5 Development IDE

VS code is used to build the project as it can handle all the required programming language and frameworks for building this project

5.6 Language of Programming

The "Mansak Amin" system leverages a combination of programming languages and frameworks tailored for real-time video processing, artificial intelligence, and interactive user interfaces. The following languages were chosen based on their suitability for specific system components:

- **Python:**

Used for AI and machine learning components, particularly for computer vision tasks (YOLOv8, OpenCV), and for backend logic including data analysis, processing video feeds, and generating alerts.

- **JavaScript (with Node.js):**

Utilized for backend services that manage real-time data flow and server-side APIs.

- **HTML/CSS and JavaScript (React.js):**

Used for building responsive and interactive front-end interfaces for system dashboards, live camera feeds, and user interaction modules.

5.7 Database

SQL / NoSQL (MongoDB):

For managing structured and unstructured data related to crowd analytics, event logs, and user management.

5.8 Web-Based Server / Other Engines

The “Mansak Amin” system is built on a web-based architecture to ensure scalability, remote access, and real-time processing capabilities. It uses several backend engines and cloud services to manage its operations efficiently:

- **Web Server:**
 - **Node.js with Express.js:**
Serves as the primary web server for handling HTTP requests, managing API routes, and enabling communication between the front-end and backend services.
- **AI & Video Processing Engine:**
 - **YOLOv8 + OpenCV (Python):**
Deployed on the backend to perform real-time object detection, crowd density estimation, and facial recognition using surveillance camera feeds.
 - **Ultralytics Framework:**
For deploying and fine-tuning pre-trained deep learning models optimized for edge performance.
- **Cloud Infrastructure:**
 - **AWS (Amazon Web Services):**
Used for scalable cloud hosting, serverless functions (via AWS Lambda), storage (S3), and GPU-backed AI model deployment (EC2 instances).

5.9 Hosting of the Project

The project can be hosted on a cloud-based hosting provider, such as Amazon Web Services (AWS), Google cloud platform (GCP). The hosting used Amazon Web Services (AWS).

5.10 Design Overview (UI/ UX)

5.10.1 Developer Perspective

From the developer's perspective, the “Mansak Amin” system is designed with a focus on modularity, scalability, and real-time performance. Developers are expected to interact with various components of the system ranging from computer vision models to web-based dashboards and backend services. Key considerations include:

- **Modular Architecture:**
 - Separation of concerns is applied by dividing the system into independent services: data ingestion, video processing, anomaly detection, user interface, and alert management.
- **Development Stack:**
 - **Frontend:** Streamlit / React (admin dashboard).
 - **Backend:** Python (FastAPI), Node.js (for microservices), integrated with RESTful APIs.
 - **Machine Learning:** YOLOv8 for detection, TensorFlow / PyTorch for model training and inference.
 - **Database:** MongoDB (for unstructured data).

- **Storage:** AWS S3 for video and image storage.
- **Development Tools & Practices:**
 - Version control with **Git** and **GitHub/GitLab**.
 - Continuous Integration/Deployment (CI/CD) via **GitHub Actions** or **Jenkins**.
 - Logging and monitoring via **Prometheus** and **Grafana** or **AWS CloudWatch**.
- **Challenges from Developer View:**
 - Optimizing performance for real-time crowd detection.
 - Balancing model accuracy with inference speed.
 - Handling large volumes of video data efficiently.
 - Ensuring system reliability during high-load periods such as Hajj days.

5.10.2 User Perspective

From the user's perspective, the "**Mansak Amin**" system provides an easy-to-use interface for managing crowd safety. Key features include:

- **Authorities** can access real-time data, receive alerts, track missing persons, and view analytical reports.
- **Pilgrims** can receive emergency alerts, access safe routes, and upload images in case of emergencies (e.g., missing persons).
- **User Interface:** Real-time dashboard, dynamic map, and missing person detection system.
- **System Accessibility:** Available on web and mobile (iOS/Android), with multi-language support and privacy protection.

5.11 Diagrams:

The following diagrams can be used to design a smart Crowd management in Hajj&Omrah:

5.11.1 Use Case

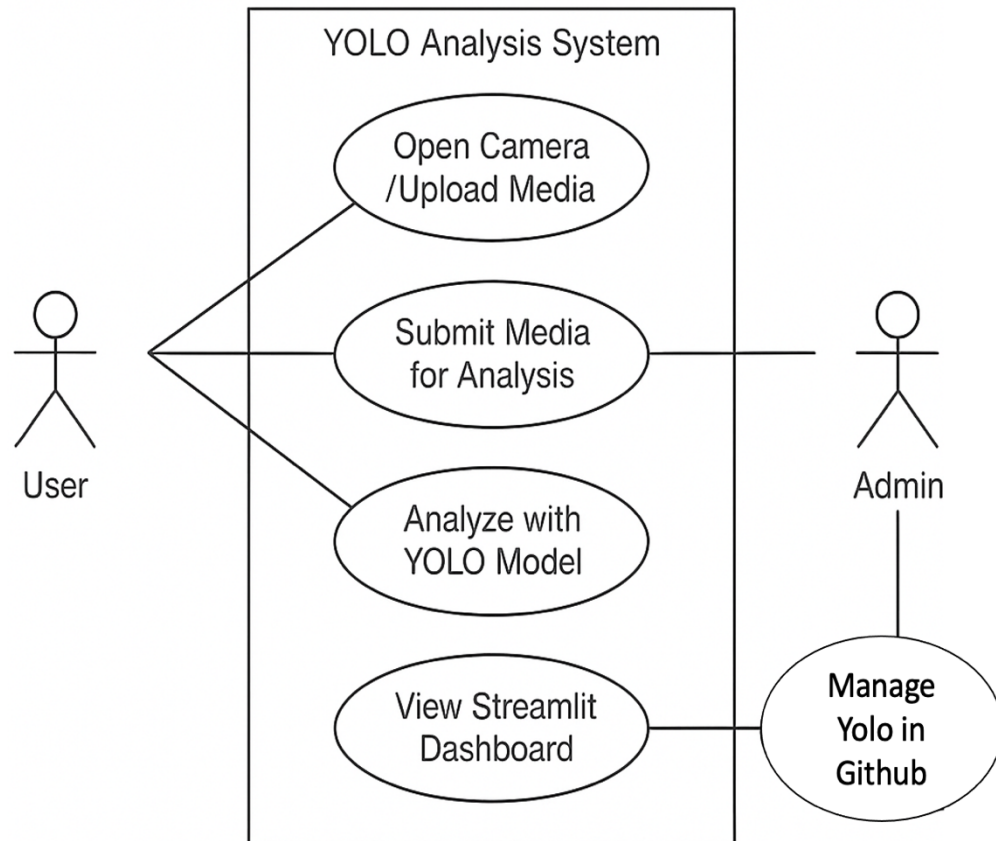


Figure 1, Use Case Diagram

Use case Description:

Name:	Actor
Brief Description:	This use case describes how student can add course to his schedule.
Goal:	Analyze a photo or video using the YOLO model.
Preconditions:	User accesses the photo analysis page.
Main Flow:	1.User opens camera or uploads media 2. Media sent to YOLO model 3. Results displayed
Alternate Flows:	If upload fails, an error message is displayed
Postconditions:	Results are shown to the user

Table 2, Use case description(Actor)

5.11.2 Iot

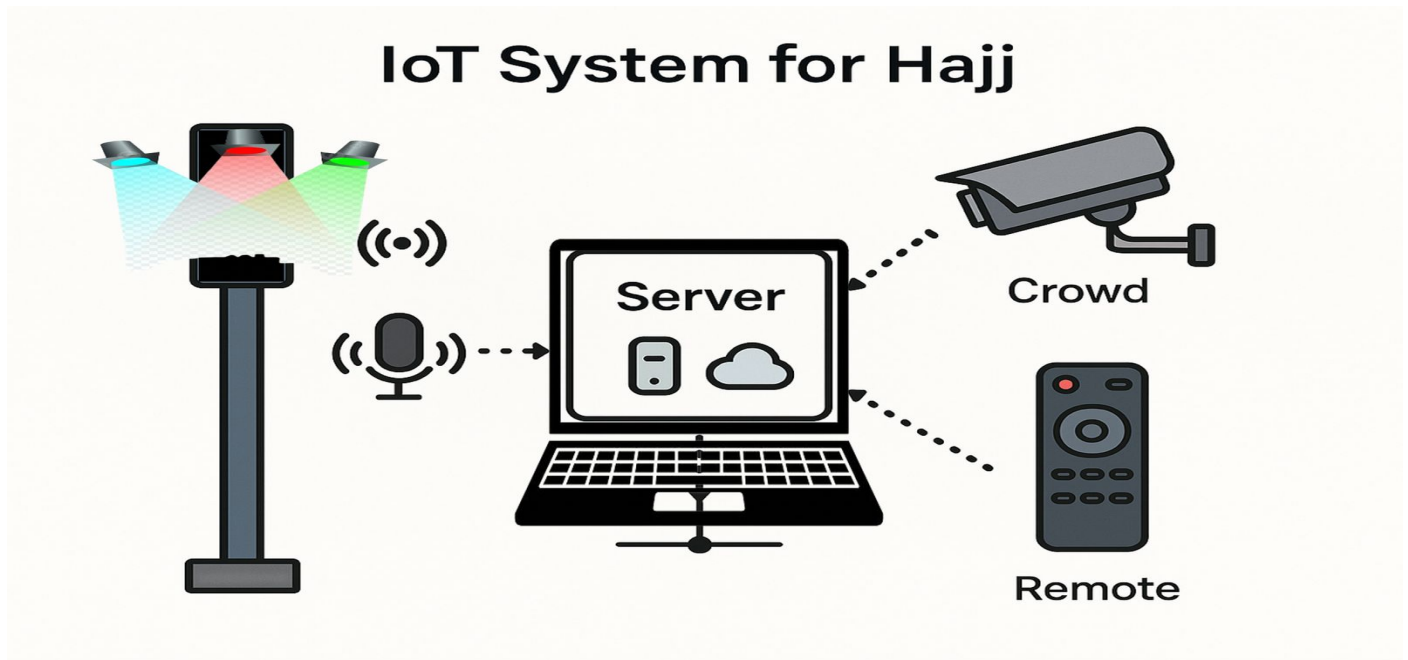


Figure 2, Iot Diagram

5.11.3 Sequence

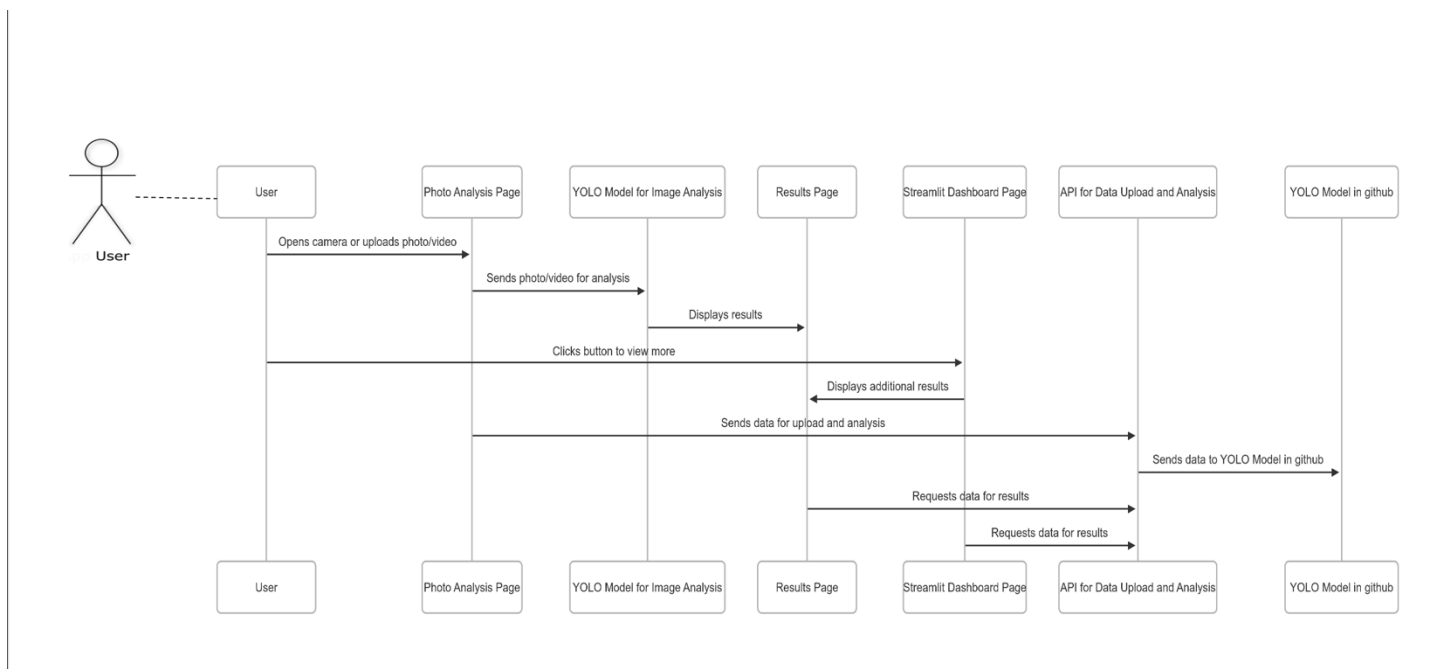


Figure 3, Sequence Diagram

5.11.4 Dataflow

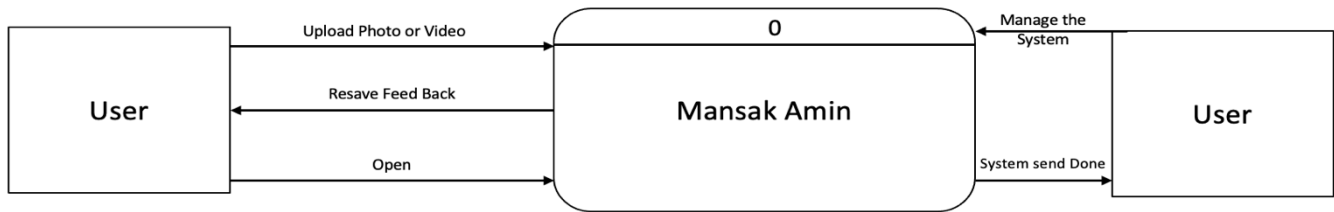


Figure 4, DFD Diagram(Context Level)

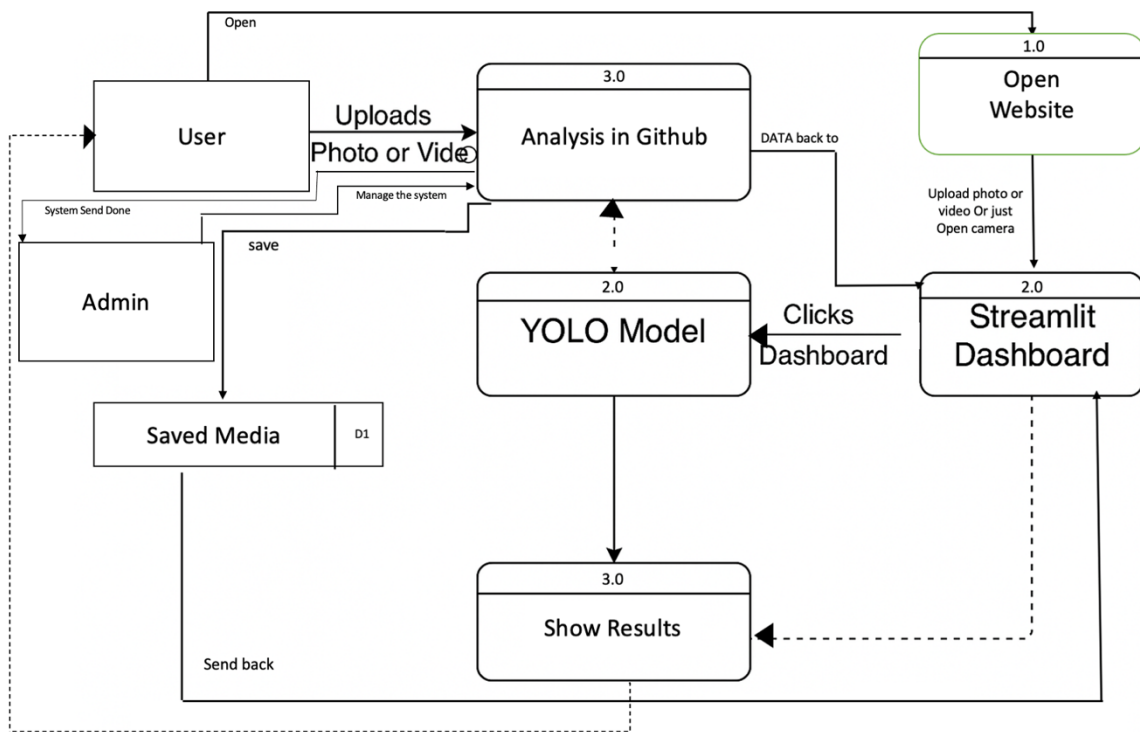


Figure 5,DFD Diagram (Level 0)

5.11.5 ER

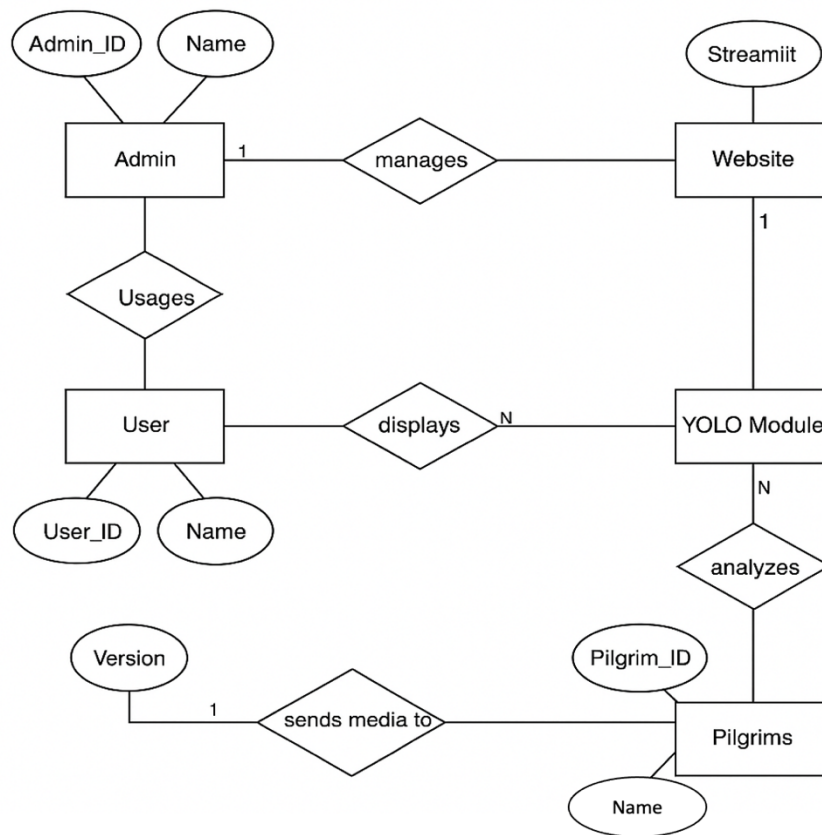


Figure 6, ER Diagram

5.11.6 Flowshart

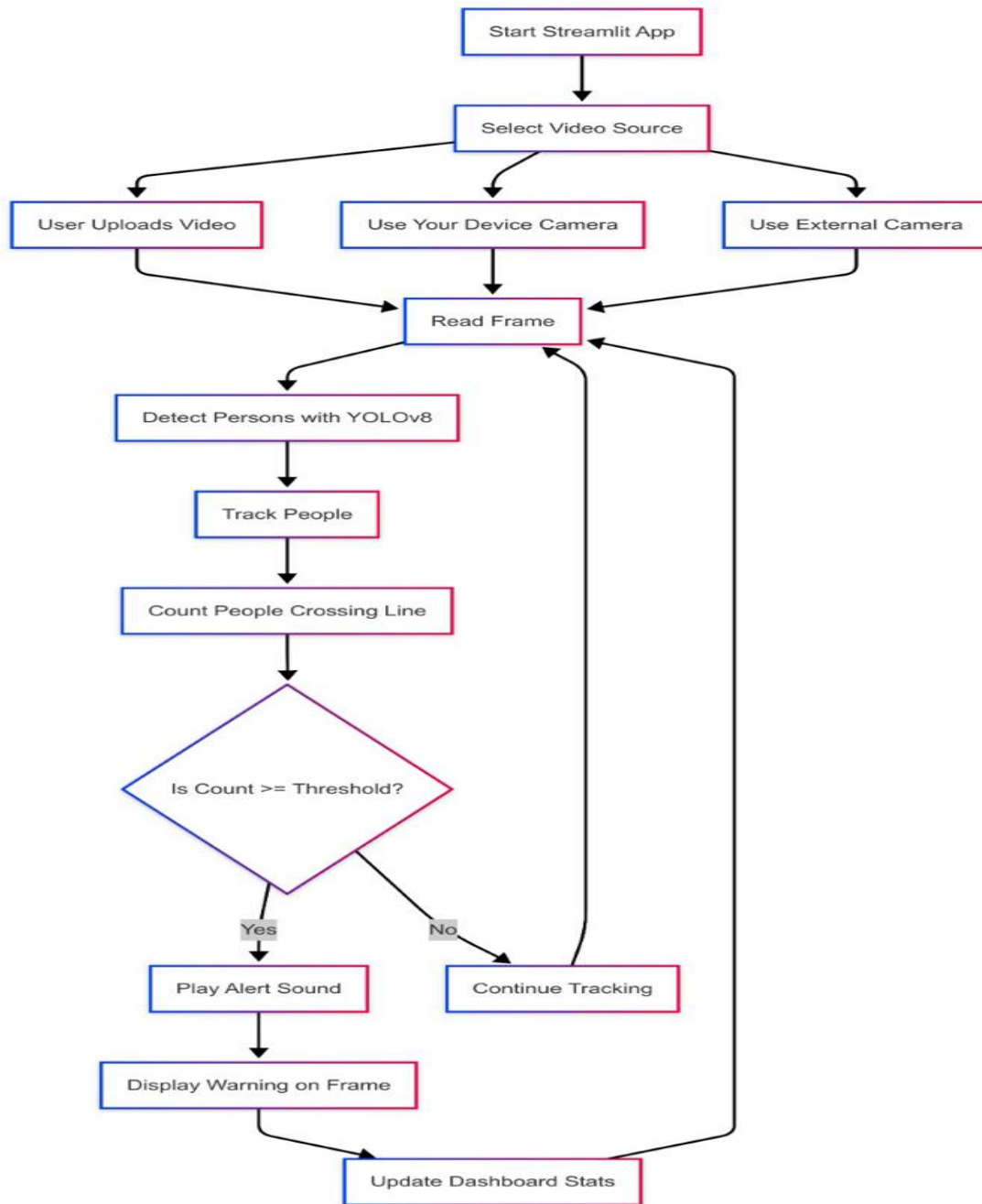


Figure 7, Flowchart

6. Optimized LSTM-Based Deep Learning Model8

YOLOv8 was released by Ultralytics on January 10th, 2023, offering cutting-edge performance in terms of accuracy and speed. Building upon the advancements of previous YOLO versions, YOLOv8 introduced new features and optimizations that make it an ideal choice for various object detection tasks in a wide range of applications.

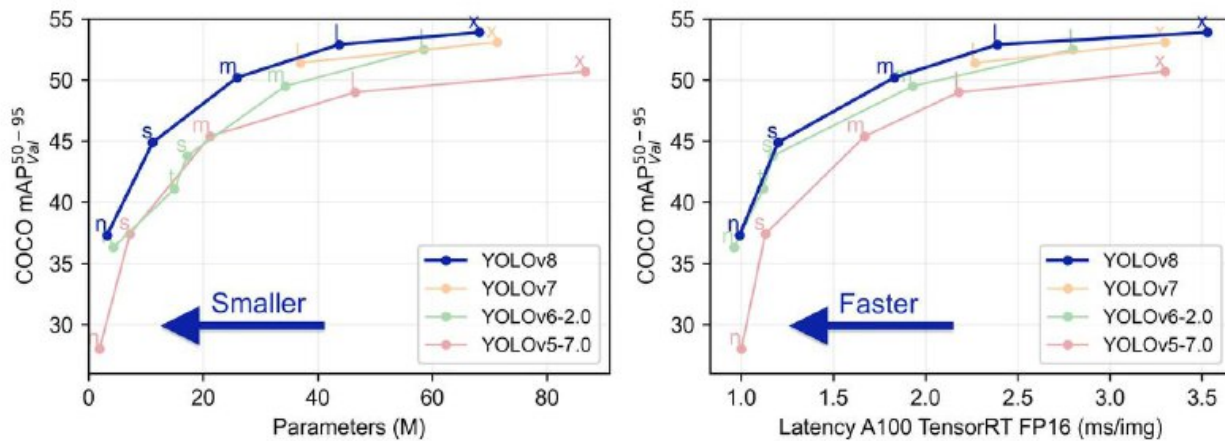


Figure 8.YOLO Module work

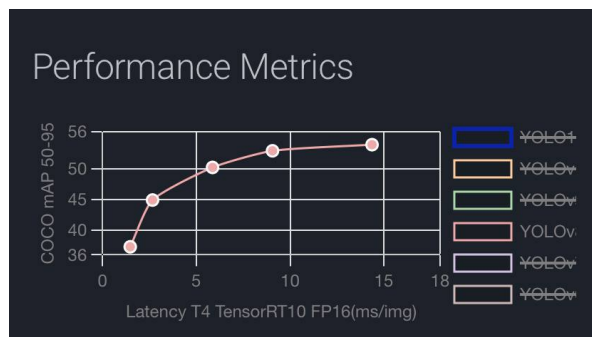


Figure 9,YOLO Module work

 Performance

Detection (COCO)

Detection (Open Images V7)

>

See [Detection Docs](#) for usage examples with these models trained on [COCO](#), which include 80 pre-trained classes.

Model	size (pixels)	mAP ^{val} 50-95	Speed CPU ONNX (ms)
YOLOv8n	640	37.3	80.4
YOLOv8s	640	44.9	128.4
YOLOv8m	640	50.2	234.7
YOLOv8l	640	52.9	375.2
YOLOv8x	640	53.9	479.1

Figure 10,YOLO Module work

7. Results and Discussion

7.1 Results

The "**Mansak Amin**" system successfully demonstrates the ability to enhance crowd management and safety during Hajj through the integration of artificial intelligence and real-time data processing. Key results include:

- **Accurate Crowd Density Analysis:** The system was able to detect and visualize crowd levels in real-time using YOLOv8 and computer vision techniques.
- **Early Warning Alerts:** The AI models effectively identified unusual crowd behavior and provided alerts to authorities in advance.
- **Missing Person Detection:** The image-matching module accurately located missing individuals by scanning live camera feeds.
- **User Interface Usability:** The dashboard provided real-time data visualization and was found intuitive and accessible to authorized users.
- **System Integration:** The platform integrated successfully with surveillance camera feeds and external data sources (e.g., weather, prayer times).

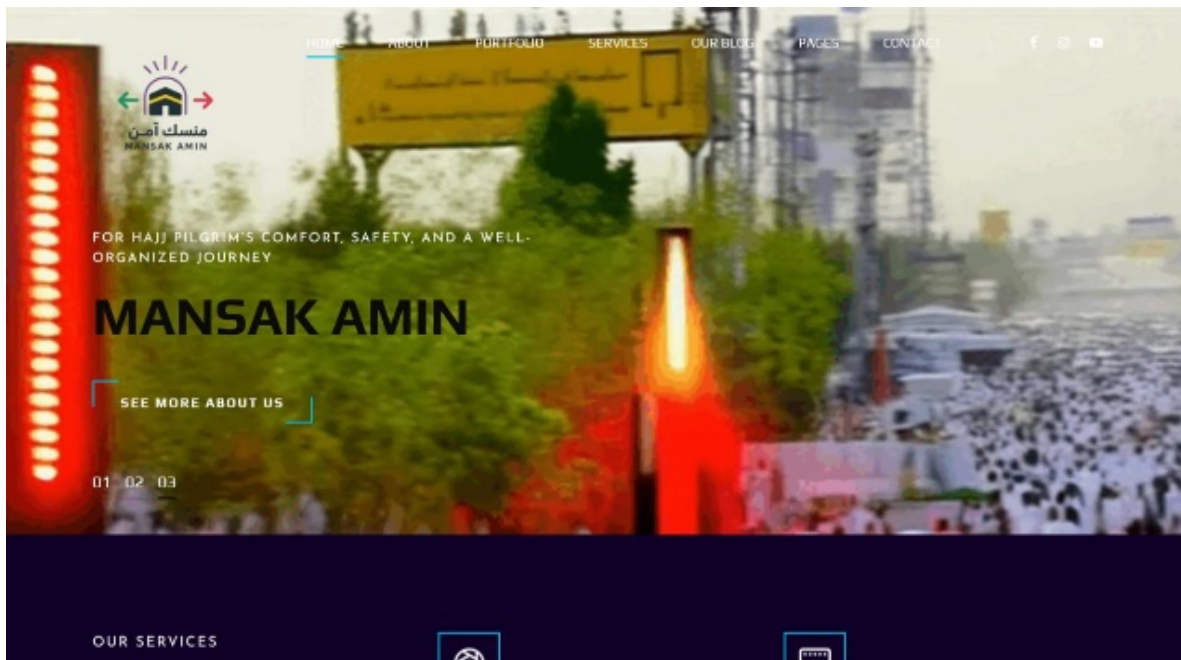


Figure 11, Mansak Amin website

4. Increased Engagement:

The "Mansak Amin" system contributes to increased engagement by enabling authorities to interact with real-time data, respond faster to potential risks, and make data-driven decisions. Its intuitive dashboard and early warning system promote proactive crowd management, which encourages a more engaged, responsive, and collaborative operational environment. Additionally, future features like drone integration and missing person detection create new levels of interaction between technology, staff, and public safety operations.

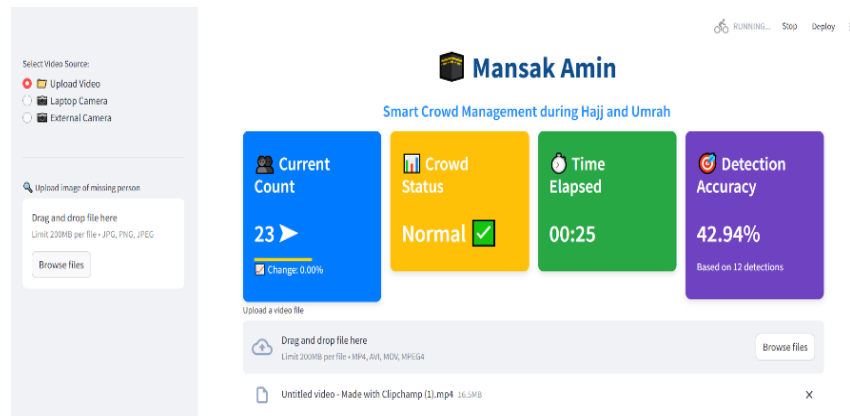


Figure 12, Dashboard Mansak Amin

5. User Engagement and Satisfaction: The system is designed with a user-centric approach, ensuring intuitive interfaces and clear visualizations that help decision-makers, security personnel, and operators easily interpret crowd conditions and respond effectively. Real-time feedback, alerts, and user-friendly controls lead to higher engagement and satisfaction. By offering reliable, accessible, and intelligent tools, "Mansak Amin" ensures users feel empowered and supported in their critical tasks.

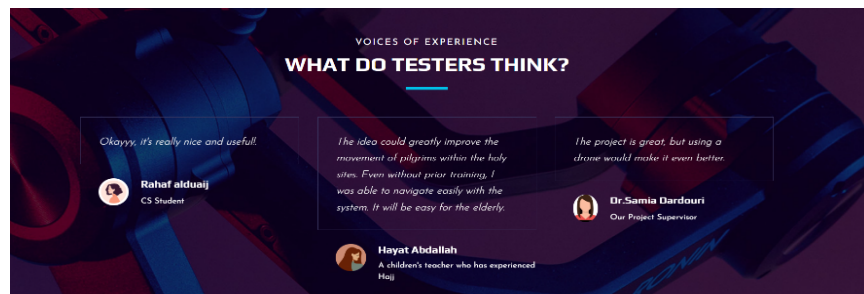


Figure 13, Study Statistic

6. Alignment with Vision 2030 Goals: The "Mansak Amin" system aligns with Vision 2030 by promoting innovation in public safety and digital transformation. It enhances the quality of services provided to pilgrims, supports smart city initiatives, and leverages AI to improve operational efficiency in crowd management. By ensuring safer and more organized Hajj experiences, the system contributes to the goal of enriching the experience of pilgrims and elevating the Kingdom's global leadership in managing large-scale religious events.

6.2 Discussion

The implementation of "Mansak Amin" highlights the practical benefits of integrating AI and computer vision in real-world crowd management scenarios. The system's ability to detect and count individuals in real time ensures timely interventions during high-density events like Hajj. However, challenges remain, such as ensuring accuracy in varying lighting conditions, avoiding false positives in detection, and managing large-scale data from multiple camera feeds. Future enhancements, including drone integration and missing person detection, will expand the system's functionality and resilience. Collaboration with security authorities and continuous system training are essential for optimizing performance and ensuring reliability during peak pilgrimage seasons.

8. Conclusions and Future Work

8.1 Conclusion:

The "Mansak Amin" system demonstrates the power of AI and computer vision in managing large crowds during Hajj. By providing real-time monitoring, alerting authorities to overcrowded areas, and guiding pilgrims using smart visual cues, the system significantly enhances safety and response efficiency. With future extensions such as drone integration and missing person detection, "Mansak Amin" is not just a reactive tool but a proactive and scalable solution aligned with Saudi Arabia's Vision 2030 goals. It paves the way for smarter, safer, and more organized pilgrimage experiences.

8.2 Future Work: Future enhancements of the "Mansak Amin" system will focus on expanding its capabilities and improving accuracy. Planned developments include integration with drone technology for aerial crowd monitoring, which allows for faster detection of congestion in wider areas. The system will also incorporate a feature for identifying and locating missing persons by analyzing uploaded photos and scanning live video feeds across all surveillance cameras. Additionally, future versions aim to support multilingual voice alerts for better communication with international pilgrims and utilize edge computing to reduce processing delays and ensure real-time responsiveness during peak hours.

References

- [1] T. Alghamdi and R. Mehmood, "Smart mobility and crowd analytics for the Hajj pilgrimage using edge AI," **Future Generation Computer Systems**, vol. 139, pp. 31–42, 2023, doi: 10.1016/j.future.2022.08.004.
- [2] R. Alharthi and O. Alfarraj, "IoT-enabled smart Hajj management system for efficient crowd control," **Sensors**, vol. 24, no.2, p. 556, 2024, doi: 10.3390/sensors24020556.
- [3] Ministry of Media, "Virtual Media Center," [Online]. Available: <https://vpc.media.gov.sa/Home/Index>. [Accessed: 28-Apr-2025].
- [4] ITAS, "How Artificial Intelligence Technologies Contributed to the Success of Hajj 2024," Jul. 2, 2024. [Online]. Available: <https://itas.sa/-/كيف-ساعدت-تقنيات-الذكاء-الاصطناعي-في-ن>. [Accessed: Apr. 28, 2025].
- [5] CNN Business Arabic, "Artificial Intelligence in Hajj 2024: A Remarkable Technological Revolution," Jun. 10, 2024. [Online]. Available: <https://cnnbusinessarabic.com/technology/66288/-2024-الذكاء-الاصطناعي-في-الحج>. [Accessed: Apr. 28, 2025].